PART VI · 30 MINUTES

Evaluation, governance and reliability under change

What evidence justifies calling a system self-improving and who decides when it may learn?

Self-improvement is as much an evaluation problem as a learning problem.

A plausible architecture is not evidence of improvement. State changes and mechanisms make the claim **testable**; they do not **settle** it.

What improved?	Relative to what?	With which resources?	For how long?	With what uncertainty?

Evaluation shifts from a snapshot to a lifecycle

Snapshot Evaluation: one model, one test set, one score

Lifecycle Evaluation: initial quality, forgetting and regressions, hallucinated updates, human review efficiency, temporal robustness, constraint satisfaction, provenance quality, time to recover from errors

Key questions:

Is the system better after a week, a month, a year?

Is the process that changes it reliable?

Declare the evaluation before you see the result

TASK: evidence-supported extraction from new documents

POPULATION: domains, sources, languages, time range

RESOURCES: inference budget, labels, review effort

STATE BOUNDARY: which components may change

METRIC: with its matching policy and judge version

COMPARISON: frozen, maintained, or a simpler remedy

Report two reference frames

$$P_{ref}(S_t) = E_{x \sim D_{ref}}[\, score(S_t, x) \,]$$

fixed reference distribution D_{ref} · operational distribution D_t at time t

Up on D_t, down on D_{ref}
Adapting while forgetting. A single moving score hides it.

Flat on D_{ref}, failing on D_t
Preserving while not adapting. A single fixed score hides it.

The expectation is a target quantity; real evaluations use finite samples and depend on budget, evaluator and environment — all of which must be specified.

Does the system stay correct as the world changes?

Does the system stay correct as the world changes?

FreshLLMs

Vu, Tu, et al. "Freshllms: Refreshing large language models with search engine augmentation." *Findings of the Association for Computational Linguistics: ACL 2024*. 2024.

FRESHLLMS: Refreshing Large Language Models with Search Engine Augmentation

Tu Vu[1] **Mohit Iyyer**[2] **Xuezhi Wang**[1] **Noah Constant**[1] **Jerry Wei**[1]
Jason Wei[3*] **Chris Tar**[1] **Yun-Hsuan Sung**[1] **Denny Zhou**[1] **Quoc Le**[1] **Thang Luong**[1]

Google[1] University of Massachusetts Amherst[2] OpenAI[3]
freshllms@google.com

Abstract

Since most large language models (LLMs) are trained once and never updated, they struggle to dynamically adapt to our ever-changing world. In this work, we present FRESHQA, a *dynamic* QA benchmark that tests a model's ability to answer questions that may require reasoning over up-to-date world knowledge. We develop a two-mode human evaluation procedure to measure both correctness and hallucination, which we use to benchmark both closed and open-source LLMs by collecting >50K human judgments. We observe that all LLMs struggle to answer questions that require encoded in their parameters. While additional training using human feedback (Ouyang et al., 2022) or knowledge-enhanced tasks can mitigate this issue, it is not easily scalable for real-time knowledge updates (e.g., stock prices). In-context learning (Brown et al., 2020) is an appealing alternative by which real-time knowledge can be injected into an LLM's prompt. While recent work has begun to explore augmenting LLM prompts with web search results (Lazaridou et al., 2022; Press et al., 2022), it is unclear how to take full advantage of search engine outputs to increase LLM factuality.

In this work, we collect FRESHQA, a novel bench-

Does the system stay correct as the world changes?

FreshLLMs

Vu, Tu, et al. "Freshllms: l
language models with sea
augmentation." *Findings o*
Association for Computat
ACL 2024. 2024.

Type	Question	Answer (as of this writing)
never-changing	*Has Virginia Woolf's novel about the Ramsay family entered the public domain in the United States?*	***Yes**, Virginia Woolf's 1927 novel To the Lighthouse entered the public domain in 2023.*
never-changing	*What breed of dog was Queen Elizabeth II of England famous for keeping?*	***Pembroke Welsh Corgi** dogs.*
slow-changing	*How many vehicle models does Tesla offer?*	*Tesla offers **six** vehicle models: Model S, Model X, Model 3, Model Y, Tesla Semi, and Cybertruck.*
slow-changing	*Which team holds the record for largest deficit overcome to win an NFL game?*	*The record for the largest NFL comeback is held by the **Minnesota Vikings**.*
fast-changing	*Which game won the Spiel des Jahres award most recently?*	***Dorfromantik** won the 2023 Spiel des Jahres.*
fast-changing	*What is Brad Pitt's most recent movie as an actor*	*Brad Pitt is credited as Keith in **IF**.*
false-premise	*What was the text of Donald Trump's first tweet in 2022, made after his unbanning from Twitter by Elon Musk?*	*He **did not tweet** in 2022.*
false-premise	*In which round did Novak Djokovic lose at the 2022 Australian Open?*	*He **was not allowed to play** at the tournament due to his vaccination status.*

Figure 1: FRESHQA exemplars. Our questions are broadly divided into *four* main categories: *never-changing*, in which the answer almost never changes; *slow-changing*, in which the answer typically changes over the course of several years; *fast-changing*, in which the answer typically changes within a year or less; and *false-premise*, which includes questions whose premises are factually incorrect and thus have to be rebutted.

Does the system stay correct as the world changes?

EvoWiki

Tang, Wei, et al. "Evowiki: Evaluating llms on evolving knowledge." *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. 2025.

EvoWiki: Evaluating LLMs on Evolving Knowledge

Wei Tang [1,2], **Yixin Cao** [3*], **Yang Deng** [4], **Jiahao Ying** [4], **Bo Wang** [5], **Yizhe Yang** [5], **Yuyue Zhao** [1,2], **Qi Zhang** [3], **Xuanjing Huang** [3], **Yugang Jiang** [3], **Yong Liao** [1,2]
[1] University of Science and Technology of China
[2] CCCD Key Lab of Ministry of Culture and Tourism
[3] Institute of Trustworthy Embodied AI, Fudan University
[4] Singapore Management University, [5] Beijing Institute of Technology
weitangcs@gmail.com yxcao@fudan.edu.cn

Abstract

Knowledge utilization is a critical aspect of LLMs, and understanding how they adapt to evolving knowledge is essential for their effective deployment. However, existing benchmarks are predominantly static, failing to capture the evolving nature of LLMs and knowledge, leading to inaccuracies and vulnerabili-

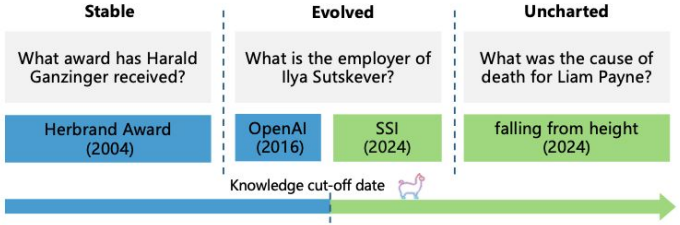


Figure 1: EvoWiki categorizes knowledge into three states according to the cut-off date of the LLMs.

Longitudinal Evaluation

Mitchell, Tom, et al. "Never-ending learning." *Communications of the ACM* 61.5 (2018): 103-115.

DOI:10.1145/3191513

Never-Ending Learning

By T. Mitchell, W. Cohen, E. Hruschka, P. Talukdar, B. Yang, J. Betteridge, A. Carlson, B. Dalvi, M. Gardner, B. Kisiel, J. Krishnamurthy, N. Lao, K. Mazaitis, T. Mohamed, N. Nakashole, E. Platanios, A. Ritter, M. Samadi, B. Settles, R. Wang, D. Wijaya, A. Gupta, X. Chen, A. Saparov, M. Greaves, and J. Welling

Abstract

Whereas people learn many different types of knowledge from diverse experiences over many years, and become better learners over time, most current machine learning systems are much more narrow, learning just a single function or data model based on statistical analysis of a single data set. We suggest that people learn better than computers precisely because of this difference, and we suggest a key direction for machine learning research is to develop software architectures that enable intelligent agents to also learn many types of knowledge, continuously over many years, and to become better learners over time. In this paper we define more precisely this *never-ending learning* paradigm for machine learning, and we present one case study: the Never-Ending Language Learner (NELL), which achieves a number of the desired properties of a never-ending learner. NELL has been learning to read the Web 24hrs/day since January 2010, and so far has acquired a knowledge base with 120mn diverse, confidence-weighted beliefs (e.g., *servedWith(tea,biscuits)*), while learning thousands of interrelated functions that continually improve its reading competence over time. NELL has also learned to reason over its knowledge base to infer new beliefs it has not yet read from those it has, and NELL is inventing new relational predicates to extend the ontology it uses to represent beliefs. We describe the design of NELL, experimental results illustrating its behavior, and discuss both its successes and shortcomings as a case study in never-ending learning. NELL can be tracked online at http://rtw.ml.cmu.edu, and followed on Twitter at @CMUNELL.

(e.g., the spam filter) given a training set of input/output pairs $\{\langle x_i, y_i \rangle\}$ of that function. Other machine learning paradigms exist as well (e.g., unsupervised clustering, topic modeling, reinforcement learning) but these paradigms also typically acquire only a single function or data model from a single dataset.

In contrast to these paradigms for learning single functions from well organized data sets over short time-frames, humans learn many different functions (i.e., different types of knowledge) over years of accumulated diverse experience, using extensive background knowledge learned from earlier experiences to guide subsequent learning. For example, humans first learn to crawl, then to walk, run, and perhaps ride a bike. They also learn to recognize objects, to predict their motions in different circumstances, and to control those motions. Importantly, they learn *cumulatively*: as they learn one thing this new knowledge helps them to more effectively learn the next, and if they revise their beliefs about the first then this change refines the second.

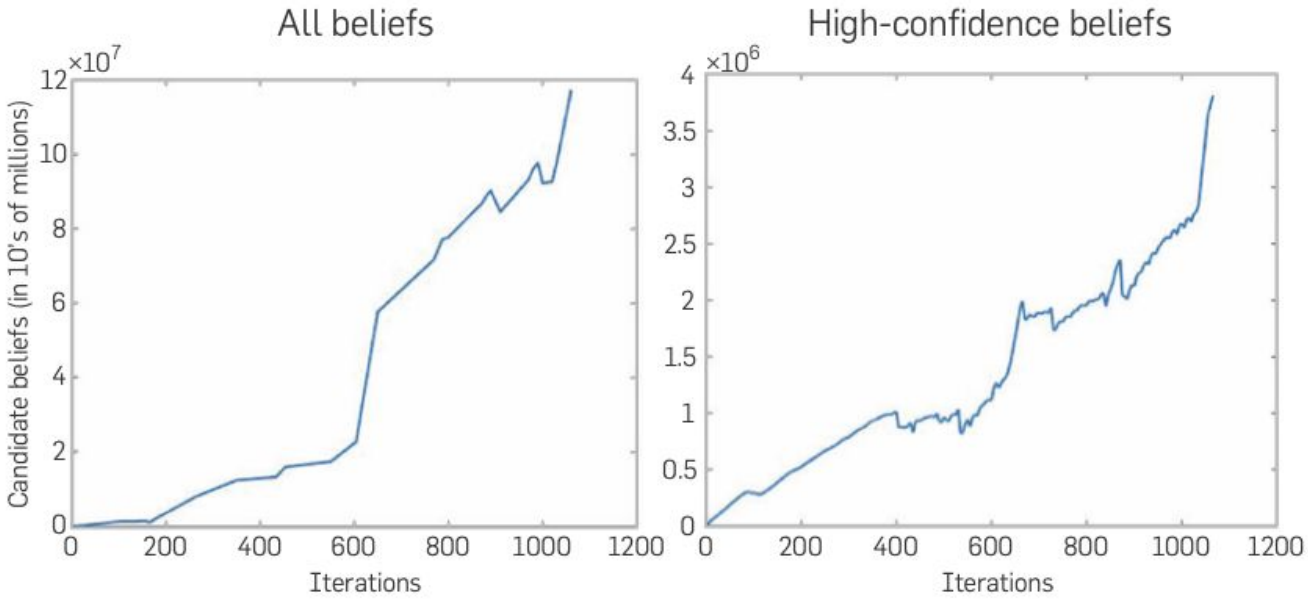
The thesis of our research is that *we will never truly understand machine or human learning until we can build computer programs that, like people,*

- learn many different types of knowledge or functions,
- from years of diverse, mostly self-supervised experience,
- in a staged curricular fashion, where previously learned knowledge enables learning further types of knowledge,
- where self-reflection and the ability to formulate new representations and new learning tasks enable the learner to avoid stagnation and performance plateaus.

Longitudinal Evaluation

Mitchell, Tom, et al. "Never-ending learning." *Communications of the ACM* 61.5 (2018): 103-115.

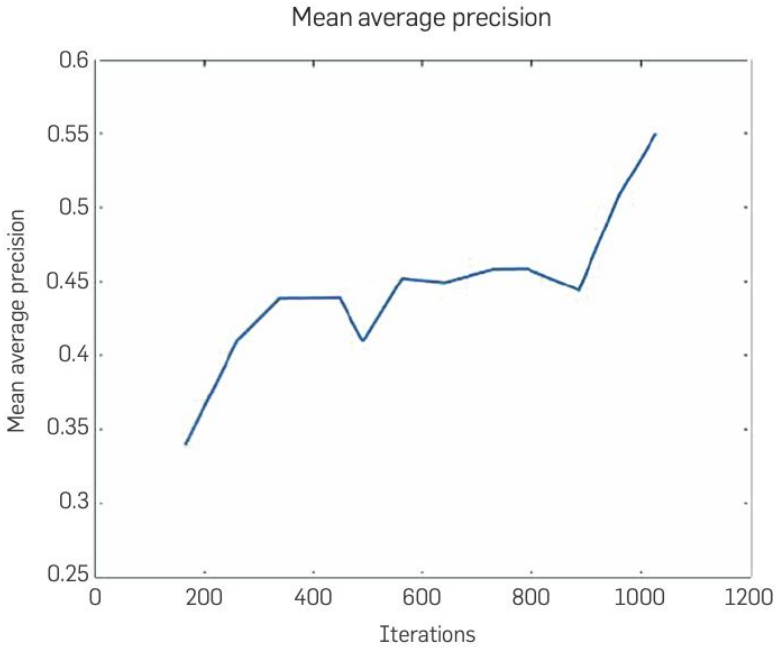
Figure 5. NELL KB size over time. Total number of beliefs (left) and number of high confidence beliefs (right) versus iterations. Left plot vertical axis is tens of millions, right plot vertical axis is in millions. The horizontal axis covers NELL iterations from January 2010 until July 2017.



Longitudinal Evaluation

Mitchell, Tom, et al. "Never-ending learning." *Communications of the ACM* 61.5 (2018): 103-115.

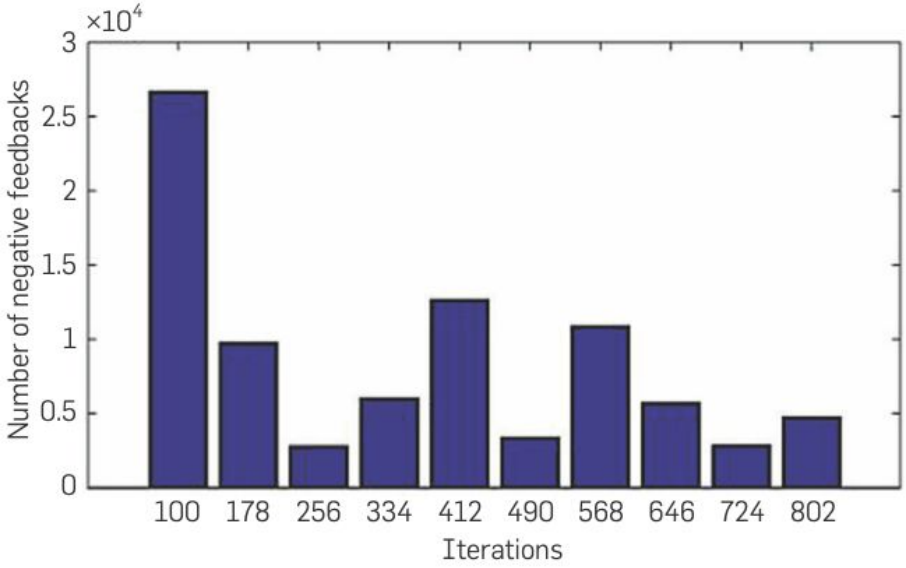
Figure 6. Evolution of NELL reading accuracy over time. The vertical axis shows the estimated Mean Average Precision over the 1000 most confident predictions for a representative sample of 18 categories and 13 relations in NELL's ontology. The horizontal axis represents NELL iterations from January 2010 through November 2016.



Longitudinal Evaluation

Mitchell, Tom, et al. "Never-ending learning." *Communications of the ACM* 61.5 (2018): 103-115.

Figure 7. Human feedback to NELL over time. Each bar in this histogram shows the number of NELL beliefs for which humans provided negative feedback, during a 78 iteration interval. This averages out to 2.4 items of feedback per month, per predicate in NELL's ontology. Human input to NELL has been dropping over time, even as its reading accuracy has been increasing.

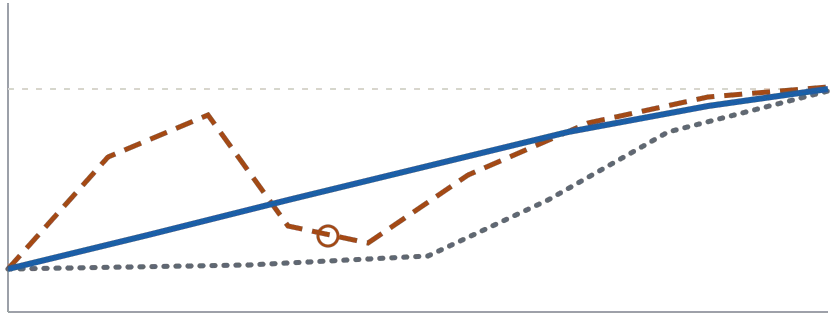


Did the update damage what used to work?

Measure

- accuracy on old tasks and on new tasks
- backward transfer — effect of new learning on old tasks
- forward transfer — effect of old learning on new tasks
- negative transfer on protected slices
- schema-level and calibration regressions

Same final score, very different learning trajectories



checkpoints over deployment time →

schematic, not results

Steady improvement

Early gain, severe regression, recovery

Late improvement

Report the path: time to benefit, worst regression, recovery time, variance across runs.

Gama et al., 2014 (concept drift and sequential evaluation)

Benchmarks test different properties

Benchmark	What it probes	Reference
LoCoMo	Very long-term conversational memory	Maharana et al., 2024
LongMemEval	Long-term interactive memory, knowledge updates, temporal reasoning	Wu et al., 2025
MTU-Bench	Multi-granularity tool use	Wang et al., 2025
AssistantBench	Realistic, time-consuming web tasks	Yoran et al., 2024
EvoWiki	Evolving knowledge	Tang et al., 2025
AgentCL	Continual learning across task streams	Shu et al., 2026
FuncBenchGen	multi-step tool-use tasks	Maekawa et al., 2026
MAWILE	workbench for auditing LLM-as-a-judge sensitivity	Hassel et al., 2026b

What a self-improvement report should show

things to report

Performance. Target performance and protected slices.

Costs. Labels, model calls, latency and review effort.

Acceptance rate. How many proposed updates were accepted.

Harm. Harmful-update rate and the worst regression.

Recovery. Time to recover after errors.

Uncertainty. Intervals and the confounds still unresolved.

Report failures too. A failed auxiliary task is evidence about the mechanism. Preservation under drift is compared with the frozen system; it is not required to rise.

Governance: when is learning allowed?

Before an update acts, four questions need answers: where it came from (provenance), who may learn what and when (policy), whether it harmed safety, fairness, calibration or old tasks (monitoring), and how to undo it and its consequences (rollback).

AgentSpec gives customizable, declarative constraints on what an LLM agent may do at run time.

Which agent components actually help, under controlled protocols.

Version thresholds and permissions: relaxing a promotion threshold alters the learning trajectory.

Learning opens a new input channel to future behaviour

- Whatever enters memory, the KB or skills shapes later answers, so each stage of the channel carries its own risk.

- Poisoning and injection can happen. Seeded false relations, instructions hidden in retrieved text, manipulated feedback. Document content is never permission to modify configuration.

- Reward hacking is a risk. Optimizing a proxy, such as an auxiliary metric or a lenient judge, instead of the target.

- Privacy leakage can also be a risk. Memories and traces hold personal data. Keep only what learning and audit require, with access and retention controls.

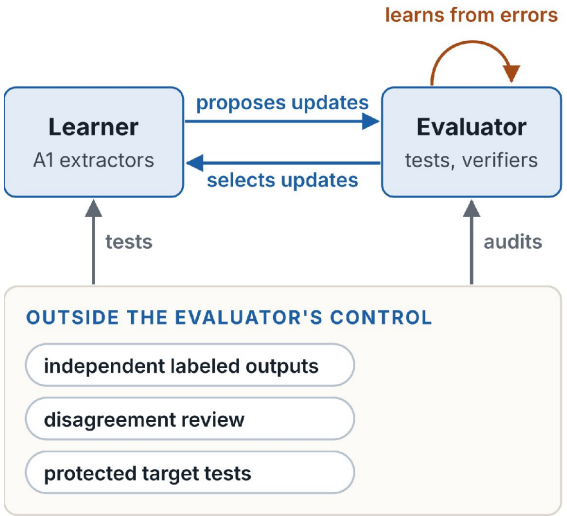
Evaluators and policies evolve too: keep them accountable

Evaluators learn. When A1 adds a test after E5 or induces a verifier, present extraction is unchanged but future update selection changes, so the evaluator is part of the learning system. Judge it by later decisions, not only current accuracy.

Never its own sole authority. A verifier that approves the examples it was induced from shows fit, not generalization. Keep independent labeled outputs, disagreement review and protected target tests outside the loop it controls.

Document the whole system. Model Cards report intended use, evaluation and limits for a model. An evolving learner needs the same record for the complete system: update permissions, evidence exposure, version bundles, known regressions and retention policy.

Source: Mitchell et al., Model Cards, FAT* 2019.



SYSTEM CARD
for the whole learning system

Update permissions
who may change what

Evidence exposure
what each part has seen

Version bundles
model, memory, evaluator, policy

Known regressions
and how they were handled

Retention policy
which traces, kept how long

cf. Model Cards, Mitchell et al., 2019

KEY MESSAGE · PART V

A self-improving system is only as trustworthy as its evaluation loop.

Show what improved, relative to what, with which resources, for how long — and with what remaining uncertainty.

Next · Part VII — Wrap-up.